

STAT-654 Project Report

Performance Analysis of Ensemble Methods for Hotel Booking Cancellation Prediction

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March 2026

1 Abstract

In this project, we analyze the Hotel Booking Demand dataset, which consists of approximately 120,000 hotel reservation records, and formulate a binary classification task to predict booking cancellations. The primary objective is to systematically compare the performance of statistical and machine learning models on this problem.

We implement a range of models, including traditional statistical approaches such as logistic regression, as well as machine learning methods such as decision trees and ensemble techniques, including random forests and gradient boosting. Hyperparameter tuning is performed using grid search to obtain optimal model configurations.

The models are evaluated across multiple dimensions, including predictive performance (accuracy, precision, recall, and F1-score) and computational efficiency (training time and model complexity). Additionally, we analyze model behavior and interpretability to understand the factors influencing performance differences, highlighting the trade-offs between statistical and machine learning approaches.

The dataset used in this study is publicly available on Kaggle.¹

¹<https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand>